Hamilton Operators of l^N Configurations

Reinhard Caspary

Central field approximation

Hamiltonian of an ion with a nucleus of charge Z and N electrons in partly filled shells:

$$H = -\sum_i \frac{p_i^2}{2m_e} - \sum_i \frac{Ze^2}{r_i} + \sum_{i<j} \frac{e^2}{r_{ij}}$$

Approximation: Every electron is located in a central potential U, caused by the coulomb energy of the nucleus and the other electrons:

$$H_0 = \sum_i \left[-\frac{p_i^2}{2m_e} + U(\vec{r}_i) \right]$$

H_0 therefore is a sum of one-electron operators.

Eigenfunctions of $\mathbf{H_0}$

The solution of the Schrödinger equation of H_0 for one electron splits into a radial and an orbital part as for the Hydrogen atom:

$$\psi_\alpha = \frac{1}{r} R_{nl}(r) Y^l_{m_l}(\theta, \phi) \chi_{m_s}$$

where $\alpha = (nlm_l m_s)$. In the case of N electrons, the radial-orbit determinant of these eigenfunctions provides an antisymetric solution of H_0, the so-called determinantal product states:

$$\Psi_0 = \frac{\det[\psi_{\alpha_j}(i)]}{\sqrt{N}} = \{\alpha_1 \alpha_2 \ldots \alpha_N\}$$

Inside a given configuration, all states Ψ_0 are completely degenerated with the energy E_0.

Perturbation theory

All interactions, which are not contained in H_0 are sumed up in the perturbation operator

$$E = E_0 + \sum_i \langle \Psi_0' | H_i | \Psi_0 \rangle$$

The radial parts of the perturbation matrix elements are treated as parameters X^i, as they depend on the unknown Potential U. The orbital part x_i of the matrix elements is calculated:

$$E = E_0 + \sum_i X^i x_i$$

Free ion hamiltonian

The free ion hamiltonian consists of operators of electric and magnetic interactions inside the ground configuration and of effective operators that take interactions with other configurations into account.

For the f^N configuration, the following radial parameters and the appropriate angular operators apear:

	Intra-configuration	Inter-configuration
electric	H_1: F^2, F^4, F^6	H_3: α, β, γ H_4: $T^2, T^3, T^4, T^6, T^7, T^8$
magnetic	H_2: ζ H_5: M^0, M^2, M^4	H_6: P^2, P^4, P^6

Coulomb interaction

The first component of the perturbation operator is the remaining part of H not contained in H_0:

$$H - H_0 = \sum_{i=1}^{N} \left[-\frac{Ze^2}{r_i} - U(\vec{r}_i) \right] + \sum_{i<j} \frac{e^2}{r_{ij}}$$

For all states of a given configuration the first part leads to the same energy shift, therefore

$$H_1 = \sum_{i<j} \frac{e^2}{r_{ij}} = e^2 \sum_{i<j} \frac{r_<^k}{r_>^{k+1}} \left(\hat{c}_i^{(k)} \cdot \hat{c}_j^{(k)} \right)$$

The energy shift is

$$E_1 = \sum_{\substack{k=2 \\ k\,\text{even}}}^{2l} F^k f_k \qquad \hat{f}_k = \sum_{i<j} \left(\hat{c}_i^{(k)} \cdot \hat{c}_j^{(k)} \right)$$

Spin-orbit interaction

The hamiltonian of the spin-orbit interaction is given by the following expression:

$$H_2 = \sum_i \frac{\hbar^2}{2m^2c^2 r_i} \frac{\mathrm{d}U(\vec{r}_i)}{\mathrm{d}r_i} (\vec{l}_i \cdot \vec{s}_i)$$

The energy shift is

$$E_2 = \zeta_{nl} \, Z_{so}$$

$$\hat{Z}_{so} = \sum_i (\vec{l}_i \cdot \vec{s}_i)$$

Coulomb configuration interaction I

The coulomb interaction of the l^N configuration with the three types of configurations

$$l^{n-2}l'^2 \,/\, l^{n-2}l'l'', \quad l'^{4l'}l^{n+2} \,/\, l'^{4l'+1}l''^{4l''+1}l^{n+2}, \text{ and } \quad l'^{4l'+1}l^n l''$$

leads to the following energy shift:

- $l = 1$: $\qquad E_3 = \alpha L(L+1)$
- $l = 2$: $\qquad E_3 = \alpha L(L+1) + \beta G(R_5)$
- $l = 3$: $\qquad E_3 = \alpha L(L+1) + \beta G(G_2) + \gamma G(R_7)$

$L(L+1)$ is the matrix element of $\vec{L}^2$, $G(S)$ is the matrix element of Casimir's operator of the symetry group S.

Coulomb configuration interaction II

The coulomb interaction of the l^N configuration with the remaining two relevant types of configurations

$$l^{n-1}l' \quad \text{and} \quad l'^{4l'+1}l^{n+1}$$

leads to the following energy shift:

$$E_4 = \sum_n T^n t_n$$

$$\hat{t}_n = 3!\, \langle kk'k''|n\rangle \sum_{i<j<k} (\hat{v}_i^{(k)} \cdot \hat{v}_j^{(k')} \cdot \hat{v}_k^{(k'')})$$

The parameters k, k', and k'' may have even values between 2 and $2l$ and their order is of no importance. Therefore n may run from 1 to $l^2 + l!/[(l-3)!3!]$. But in case of f^N configurations, only the T^n's with $n = 2, 3, 4, 6, 7, 8$ are lineary independant.

Spin-spin and spin-other-orbit interaction

The hamiltonian of the spin-spin and spin-other-orbit interaction is given by the expression

$$H_5 = \frac{e^2\hbar^2}{m^2c^2}\sum_{i<j}\left[\frac{(\vec{s}_i\cdot\vec{s}_i)}{\vec{r}_{ij}^{\,3}} - \frac{3(\vec{r}_{ij}\cdot\vec{s}_i)(\vec{r}_{ij}\cdot\vec{s}_j)}{r_{ij}^5} + \left([\vec{\nabla}_i\frac{1}{r_{ij}}\times\vec{p}_i]\cdot[\vec{s}_i+2\vec{s}_j]\right)\right]$$

The energy shift is

$$E_5 = \sum_{\substack{k=0\\ k\,\text{even}}}^{2l-2} M^k m_k$$

Operator $\hat{\mathsf{m}}_\mathsf{k}$

$$\hat{m}_k = \sum_{i\neq j} \Bigg[a \Big[\{\hat{w}_i^{(0,k)} \times \hat{w}_j^{(1,k+1)}\}^{(11)0} + 2\,\{\hat{w}_i^{(0,k+1)} \times \hat{w}_j^{(1,k)}\}^{(11)0} \Big]$$

$$+ b \Big[\{\hat{w}_i^{(0,k+2)} \times \hat{w}_j^{(1,k+1)}\}^{(11)0} + 2\,\{\hat{w}_i^{(0,k+1)} \times \hat{w}_j^{(1,k+2)}\}^{(11)0} \Big]$$

$$+ c\,\{\hat{w}_i^{(1,k)} \times \hat{w}_j^{(1,k+2)}\}^{(22)0} \Bigg]$$

Judd 1968: (1), (2)

$$a = \langle l || \hat{C}^{(k)} || l \rangle^2 \sqrt{(k+1)(2l+k+2)(2l-k)}$$

$$b = \langle l || \hat{C}^{(k+2)} || l \rangle^2 \sqrt{(k+2)(2l+k+3)(2l-k-1)}$$

$$c = -2 \langle l || \hat{C}^{(k)} || l \rangle \, \langle l || \hat{C}^{(k+2)} || l \rangle \sqrt{(k+1)(k+2)(2k+3)}$$

Spin-orbit configuration interaction

The spin-orbit interaction of the $(nl)^N$ configuration with the three configurations

$$(nl)^{N-1}n'l\,, \quad (n'l)^{4l+1}(nl)^{N+1}\,, \text{ and } \quad (n'l')^{4l'+1}(nl)^N n''l'$$

leads to the energy shift

$$E_6 = \sum_{\substack{k=2 \\ k \text{ even}}}^{2l} P^k p_k$$

Operator $\hat{\mathbf{p}}_{\mathbf{k}}$

$$\hat{p}_k = \langle l||\hat{C}^{(k)}||l\rangle^2 \frac{1}{\sqrt{2(2k+1)}}$$

$$\times \sum_{i\neq j} \left[\sqrt{\frac{(2l+k+2)(2l-k)(k+1)}{2k+3}} \; \{\hat{w}_i^{(0,k)} \times \hat{w}_j^{(1,k+1)}\}^{(11)0} \right.$$

$$\left. - \sqrt{\frac{(2l+k+1)(2l-k+1)k}{2k+1}} \; \{\hat{w}_i^{(0,k)} \times \hat{w}_j^{(1,k-1)}\}^{(11)0} \right]$$

Goldschmidt 1978: (1.156)

Crystal field interaction

The coulomb interaction of the electrons with an external charge distribution $\varrho(\vec{r})$ is given by

$$H_7 = -e\,\varrho(\vec{r}) \sum_i \sum_{k,q} \frac{r_i^k}{r^{k+1}} \sqrt{\frac{4\pi}{2k+1}} Y_q^k(\theta,\varphi) (C_q^{(k)})_i$$

The energy shift is

$$E_7 = \sum_{\substack{k=2 \\ k \text{ even}}}^{2l} \sum_{q \text{ even}} B_q^k \sum_i (C_q^{(k)})_i$$

$$B_{-q}^k = (-1)^q B_q^k$$

Magnetic field interaction

The hamiltonian of an external magnetic field is

$$H_8 = \mu \left(\vec{B} \cdot \sum_i [\vec{l}_i + g_s \vec{s}_i] \right)$$

The energy shift is

$$E_8 = B_z Z_m$$

$$\widehat{Z}_m = \mu \sum_i [(l_z)_i + g_s (s_z)_i]$$